



DMMMSU

SOUTH LA UNION CAMPUS
COLLEGE OF COMPUTER SCIENCE

GUIDE TO THESIS WRITING

FIRST EDITION

ISBN 978-971-0174-91-1

**COLLEGE OF COMPUTER SCIENCE
GUIDE IN THESIS WRITING (First Edition)**

CCS RESEARCH FORMAT TECHNICAL WORKING GROUP

**DURAN, ISIDRO L.
MARZAN, CHARLIE S.
MUNAR, EMELY A.
SONGCUAN, JEROME P.
VERAS JR., AGUSTIN R.**

**DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
SOUTH LA UNION CAMPUS
AGOO, LA UNION**

COLLEGE OF COMPUTER SCIENCE

2016

Part 1

GENERAL FORMAT

General Format and Style in Writing Research Report

1. The manuscript must be encoded in a short bond paper (8 ½ x 11).
2. The margins are as follows:

a. Top	:	1.3 inches
b. Bottom	:	1 inch
c. Left	:	1.5 inches
d. Right	:	1 inch

Do not use any borderline on the margins.

3. Font style to be used is Times New Roman, font size 12.
4. Generally, follow double spacing in encoding the manuscript.
5. Type one (1) space after comma/ semi-colon; two (2) spaces after period/colon.
6. Do not abbreviate or contract words. (e.g. use *do not* instead of *don't*; *national* instead of *nat'l*).
7. Case Styles
 - a. THIS LINE IS WRITTEN IN UPPERCASE
 - b. this line is written in lower case
 - c. This Line is Written in Title Case, that is, the First Letter of Every Word is Written in Uppercase Except for Function Words (e.g. articles, prepositions)
 - d. This line is written as Sentence case, that is, only the first letter of the first word capitalized with the exceptions of Proper Nouns.
8. Alignment

This is a flush left alignment

This is a center alignment

This is a flush right alignment

The research manuscript is generally encoded in a "Justified" format.

9. Chapter Number is written in Title Case, Hindu-Arabic numeral. Chapter Title is written in UPPERCASE. Both are written in boldface and follow a center alignment.
10. All major parts of the research should appear as side headings (flush to the left margin). Sub-parts are indented.
11. Indention

- a. Set an automatic 0.5" TAB (1/2 inch) for regular paragraph indention
[0.5"] XXX
XX
- b. Use parallel indention for Table Titles

Table 1. XXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XX XXX XXXXXXXXXXXXXXXXXXXXXXXX

c. Use hanging indentation for References (Press CTRL T)

XX
[0.5"] XXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

d. Use set-off indentation for long quotations (more than 40 words)

Set-off quotations shall not use quotation marks. The quoted item is written single spaced. Set a 0.5 inch space-margin from the left margin.

[0.5"] XXX
XXXXXXXXXXXXXXXXXXXX

0.5" } XXX
XX
XX
XX (Author, year, p. 3)

12. There shall be **no** introductory statement per chapter. Each chapter starts immediately with the first sub-part of the chapter.

13. Pagination

- a. *Automatically set the page numbers of the manuscript*
- b. *Title and Start of Chapter.* The page numbers in the title page and start of every chapter are omitted/hidden (although the pagination is reflected in the Table of Contents).
- c. *Preliminaries:* Pagination in the Preliminary Sections appears at the bottom of the page using the lower case Roman numeral (i, ii, iii, iv...).
- d. *Main Body.* Pagination in the main body appears in the upper right hand corner of the page using the Hindu-Arabic numeral (1, 2, 3...)

14. For multi-level headings, follow the presentation below:

First Level Heading (boldface, flushed to the left margin)

Second Level Heading (boldface, indented half inch (TAB 0.5") from the left margin)

Third Level Heading (boldface, italicized, indented half inch (TAB 0.5") from left margin)

15. Four (4) single spaces separate the Chapter Title and the first entry heading.

16. Four (4) single spaces separate the major parts of a chapter (first level headings).

17. Double space separates minor-parts (second and third level headings).

18. For headings and titles (tables, figures, graphs, plates) with two lines or more, proper division of phrases must be observed. **Avoid** hanging adjectives and prepositions.

Wrong: Correlation Analysis Between the Profile of the
Public Safety Officers and the Officers' Assessment on the
Quality of PPSC Education and Training

Correct: Correlation Analysis Between the Profile of the Public Safety Officers
and the Officers' Assessment on the Quality of PPSC Training

Sample of Multilevel Heading and Typing Instruction

Chapter 1 (2 spaces) INTRODUCTION

(4 single spaces) Situation Analysis

Start typing here for first level heading and continue typing through the right margin and the text should automatically flow down in the set left margin using the "justified" format.

Double space separates the lines and paragraphs. Quadruple space separates the main parts of the research.

(4 single spaces) Framework of the Study

Start typing here for **first level heading**. Double space separates the sub-headings.

For sub-levels, the sub-headings must be separated as shown below. Double space separates the sub-headings.

Web Counts

Start typing here for **second level heading**.

Literal Queries. For **third level heading**, start typing here immediately after the period.

(4 single spaces) Statement of the Problem

In-Text Citation Style (Source: APA Style from Purdue OWL Online Writing Lab, 2015)

All sources that are cited in the text must appear in the reference list at the end of the paper. In-text citations must include the author and year of publication (e.g. Reyes, 2009).

As a general rule, use the **past tense** or **present perfect tense** when using signal phrases to describe earlier research. For example:

Reyes (2009) **found** or Reyes (2009) **has found**...

In-text citation capitalization, quotes, and italics/underlining

- Always capitalize proper nouns, including author names and initials: D. Reyes.
- If you refer to the title of a source within your paper, capitalize all words that are four letters long or greater within the title of a source: *Sense and Sensibilities*. Exceptions apply to short words that are verbs, nouns, pronouns, adjectives, and adverbs: *New York, To Be or Not To Be*.
(Note: in your References list, only the first word of a title will be capitalized: *To be or not to be*.)
- When capitalizing titles, capitalize both words in a hyphenated compound word: *Natural-Born Cyborgs*.
- Capitalize the first word after a dash or colon: "Defining Film Rhetoric: The Case of Hitchcock's *Vertigo*".
- Italicize or underline the titles of longer works such as books, edited collections, movies, television series, documentaries, or albums: *The Closing of the American Mind*; *The Wizard of Oz*; *Friends*.
- Put quotation marks around the titles of shorter works such as journal articles, articles from edited collections, television series episodes, and song titles: "Multimedia Narration: Constructing Possible Worlds"; "The One Where Chandler Can't Cry."

Short quotations

When directly quoting from a work, include the author, year of publication, and the page number for the reference (preceded by "p."). Introduce the quotation with a signal phrase that includes the author's last name followed by the date of publication in parentheses. Use double quotation marks (" ") when quoting texts verbatim. Use single quotation marks (' ') to highlight words within a quoted statement. To retain only the essential ideas in a long text, ellipsis (three dots) can be used to cut very long statements. The end quotation mark follows comma, period or other closing punctuation (text ,,..."). The page number is located outside the quotation marks, and enclosed in parentheses. No period follows the page number.

Fontanilla (2015) said that: "Transfer sort, as the proposed algorithm requires $O(n)$ phases when sorting reversed samples and sorted samples however it requires $O(n \log n)$ phases when sorting andom samples and $O(n^2)$ when sorpting almost reverse samples." (p. 34)

Anent to this, Donaldson (2010) said that there is a need to train learners how to participate in the interpersonal mode of thinking and linguistic expression that are important parts of cultural heritage. He said: "People who cannot handle this type of language are at a 'gross disadvantage' in every field of study..." (p.117)

If the author is not named in a signal phrase, place the author's last name, the year of publication, and the page number in parentheses after the quotation.

Large data sets such as those obtained from the web are not the panacea that are claimed to be. Rather web-based models should be used as a baseline for NLP tasks. (Lapata&keller, 2005, p.26)

Long quotations

Place direct quotations that are 40 words, or longer, in a free-standing block of typewritten lines. Start the quotation on a new line indented 1/2 inch from the left margin, i.e., in the same place you would begin a new paragraph. Type the entire quotation on the new margin, and indent the first line of any subsequent paragraph within the quotation 1/2 inch from the new margin. Do not use quotation marks. The parenthetical citation comes after the closing punctuation mark, as shown below:

Atwell (1999) illustrates how much deflation may be necessary:

0.5" [...] it turned out that one of the main things that English language teachers asked for was a 'conversation practice machine': a computer system with which a learner could 'chat' in ordinary English to practice their everyday conversational English. [...] Another thing which ELT users seemed to want was the ability to correct errors and give guidance on the case of errors in learners' conversations. (pp.31-32)

Summary or paraphrase

When referring to an idea from another work, but **NOT** directly quoting the material, or making reference to an entire book, article or other work, you only have to make reference to the **author and year of publication** and **not** the page number in your in-text reference. Quotation marks are not needed for paraphrased, summarized ideas.

Source Citation

Primary Source Citation

Primary source citation includes the author and year

For all tasks, the web-based models are compared against identical models whose parameters are estimated from the BNC (Burnard, 1995).

Secondary Source Citation

If you use a source that was cited in another source, name the original source in your signal phrase. List the secondary source in your reference list and include the secondary source in the parentheses. The format is: **(as cited in author, year)**. Take note that only the **secondary source** is given the author-year citation.

According to Lapata and Keller (as cited in Knight & Chander, 1994), the best Google model reached an accuracy of 95.51 percent. This compares favorably with the 95-96 percent upper bound.

The best Google model reached an accuracy of 95.51 percent. This compares favorably with the 95-96 percent upper bound (Lapata & Keller as cited in Knight & Chander, 1994).

Citing Author(s)

A Work by Two Authors: Name both authors in the signal phrase or in the parentheses each time you cite the work. Use the word "and" between the authors' names within the text and use the **ampersand (&)** in the parentheses.

Analogous to the argument raised by Ventola (1987) and Bhatia (1993), the notion of subgenres is different in terms of their communicative purposes and the different strategies writers use to accomplish these purposes (as cited in Berkenkotter & Huckin, 1995).

A Work by Three to Five Authors: List all the authors in the signal phrase or in parentheses the first time you cite the source.

(Kernis, Cornell, Sun, Berry, & Harlow, 1993)

In subsequent citations, only use the first author's last name followed by "et al." in the signal phrase or in parentheses.

(Kernis et al., 1993)

In *et al.*, *et* should **not** be followed by a period.

Six or More Authors: Use the first author's name followed by et al. in the signal phrase or in parentheses.

Harris et al. (2001) argued...

(Harris et al., 2001)

Unknown Author: If the work does not have an author, cite the source by its title in the signal phrase or use the first word or two in the parentheses. Titles of books and reports are italicized or underlined; titles of articles, chapters, and web pages are in quotation marks. A similar study was done of students learning to format research papers ("Using APA," 2001).

Note: In the rare case, the "Anonymous" is used for the author, treat it as the author's name (Anonymous, 2001). In the reference list, use the name Anonymous as the author.

Organization as an Author: If the author is an organization or a government agency, mention the organization in the signal phrase or in the parenthetical citation the first time you cite the source.

According to the American Psychological Association (2000),...

If the organization has a well-known abbreviation, include the abbreviation in brackets the first time the source is cited and then use only the abbreviation in later citations.

First citation: (Association of Computing Machinery [ACM], 2000)
Second citation: (ACM, 2000)

Two or More Works in the Same Parentheses: For similar ideas from different authors, citation of idea is preferred in order to avoid long individual citations discussing the same topic.

When your parenthetical citation includes two or more works, order them the same way they appear in the reference list, separated by a semi-colon.

A number of studies have investigated the usefulness of the web for word sense disambiguation (Mihalcea & Moldovan, 1999; Rigau et al., 2002; Santamaria et al., 2003), question answering (Dumais et al., 2002; Hildebrandt et al., 2004; Soricut & Brill, 2004), and language modeling (Zhu & Rosenfeld, 2001; Keller & Lapata, 2003; Bulyko et al., 2003).

Authors With the Same Last Name: To prevent confusion, use first initials with the last names.

(E. Johnson, 2001; L. Johnson, 1998)

Two or More Works by the Same Author in the Same Year: If you have two sources by the same author in the same year, use lower-case letters (a, b, c) with the year to order the entries in the reference list. Use the lower-case letters with the year in the in-text citation.

Research by Berndt (1981a) illustrated that...

In case of the same author with an important idea cited in several publications, indicate the year of publications arranged chronologically.

Theoretically, the Systemic-Functional conception of language (developed by Halliday, 1975; 1978; 1994) espouses the role of language in the social construction of life.

Introductions, Prefaces, Forewords, and Afterwords: When citing an Introduction, Preface, Foreword, or Afterwords in-text, cite the appropriate author and year as usual.

(Funk & Kolln, 1992)

Personal Communication: For interviews, letters, e-mails, and other person-to-person communication, cite the communicator's name, the fact that it was personal communication, and the date of the communication. Do not include personal communication in the reference list.

(E. Robbins, personal communication, January 4, 2001).

A. P. Smith also claimed that many of her students had difficulties with APA style (personal communication, November 3, 2002).

Electronic Sources

If possible, cite an electronic document the same as any other document by using the author-date style.

Kenneth (2000) explained...

Unknown Author and Unknown Date: If no author or date is given, use the title in your signal phrase or the first word or two of the title in the parentheses and use the abbreviation "n.d." (for "no date").

Another study of students and research decisions discovered that students succeeded with tutoring ("Tutoring and APA," n.d.).

Sources Without Page Numbers

When an electronic source lacks page numbers, you should try to include information that will help readers find the passage being cited. When an electronic document has numbered paragraphs, use the abbreviation "para." followed by the paragraph number (Hall, 2001, para. 5). If the paragraphs are not numbered and the document includes headings, provide the appropriate heading and specify the paragraph under that heading. Note that in some electronic sources, like Web pages, people can use the Find function in their browser to locate any passages you cite.

According to Smith (1997), ... (Mind over Matter section, para. 6).

Note: Never use the page numbers of Web pages you print out; different computers print Web pages with different pagination.

Other Styles

1. Use the full word in the text and Latin abbreviations inside parentheses or tables.

The Systemic-Functional model posits that language is *systemic*, **that is**, it offers systems of choices (e.g., the mood system) in language. These are reflected through register (i.e., lexicogrammatical transitivity) and subsequently gives rise to the generic text typology (Christie, 1999; Ventola, 1987).

2. Cross-referencing must be used to guide readers, especially when the table/figure/chart does not appear on the same page.

Table 3 (on page 25) shows the number of phases of each algorithm in different number of elements in reverse samples.

3. For acronyms, state the full term during first mention of the word and parenthetical explanation for the use of the abbreviation/acronym in subsequent mentioning of the term.

Previous work demonstrated that web counts can be used to approximate bigram counts, thus suggesting that web-based frequencies should be useful for a wide variety of (Natural Language Processing (NLP) tasks.

4. Use transition markers for smoother flow of ideas and make a more cohesive presentation of ideas.

Meanwhile, researches on Natural Language Processing have shown the relevance of [...]. **Accordingly**, the web-based models for NLP [...].

5. Use words for numbers 1-9. Use figures for 10 and above, except if it is the start of a sentence.
6. Use word "percent" in textual presentation and the figure (%) in parentheses and tabular/graphical presentation.

About 90 percent of the respondents belong to the low-income bracket while only few respondents are in the high-income group (7%).

7. Observe parallel indentation for lists or bulleted presentation.

Corollary to this, the following are some of the mapped taxonomy of hedges listed by Salager-Meyer (1993):

1. **Modal auxiliary verbs** (the most straightforward and widely used means of expressing modality in English academic writing), the most tentative ones being: *may, might, can, could, would, should*:
2. **Adjectival, adverbial and nominal modal phrases**:
 - 2.1. probability adjectives: e.g., *possible, probable, un/likely*
 - 2.2. nouns: e.g., *assumption, claim, possibility, estimate, suggestion*
 - 2.3. adverbs (which could be considered as non-verbal modals): e.g., *perhaps, possibly, probably, practically, likely, presumably, virtually*. (p. 10)

8. Tenses, Voice, and Point of View

In general, foreground the research using the third person point of view. (e.. "The results indicate ... "). Avoid using the "I", "we"

Generally, observe the use of the following tenses/voices of the verb:

Introduction

Situation Analysis : present/past tense/present perfect
Framework of the Study: present/past tense/present perfect
Problem : past/present tense

Methodology : past tense

Discussion : present/past tense present/ past tense/
active/passive voice

Table 2 presents the distribution of respondents...

The respondents rated the materials as highly acceptable based on...
This means that the materials can be used in the schools.

Most of the respondents at the time of the study are aged 35-40.

The data revealed that...

This result suggests that...

It was found out that...

Summary /Conclusions : past tense

Recommendation : simple present/future tense/modals

The developed materials may be offered to...

Teachers should be provided with...

Further studies on... are encouraged.

Part 2

THE PRELIMINARY SECTION

THE PRELIMINARY SECTION

The following compose the preliminary section of the manuscript:

1. COVER PAGE
2. TITLE PAGE
3. APPROVAL SHEET
4. ACKNOWLEDGMENTS
5. DEDICATION (optional)
6. TABLE OF CONTENTS
7. LIST OF TABLES
8. LIST OF FIGURES
9. LIST OF PLATES
10. ABSTRACT

The Cover Page and Title Page

The Cover and Title pages are written using Times New Roman, font size 12. The entries are written single spaced and the parts are evenly spaced.

Title

- 1) The title must be written in upper case letters; boldface, follows an inverted pyramid and a center alignment format, and shall **not** exceed three typewritten (3) lines. However, the lines must follow proper division of phrases (i.e., no instances of hanging prepositions or adjectives).
- 2) Scientific names are italicized and enclosed in parentheses.

Author(s)

- 1) Authorship may be by individual or group. For group study, there shall be a maximum of five (5) researchers to compose a group.
- 2) For group research, the name of the lead researcher shall appear first, followed by the names of the other members arranged in alphabetical order (based on surname).
- 3) The names shall be written in upper case letters and follow center alignment.
- 4) The order of presentation is: First Name, Middle Initial, Family Name
- 5) Group/Team research is not allowed in the graduate level.

Name of the University, Campus, College, Address

All names referring to the institution are written in upper case letters, center alignment.

Degree is written in upper case letters

Major/Specialization is written in Title Case

Date of Completion includes the month and year of the end of term (March 2013 or May 2013 or October 2013)

Spine

1) The entries in the spine include:

- a. Author(s) - For group research: Lead Author, et al.
- b. Title
- c. Year

2) For long titles, a running/abridged title may be used.

3) If space is limited, scientific names may be deleted in the spine.

FONTANILLA, KRANZ NIKOLAI V., et al.	TRANSFER SORT: A PROPOSED SORTING ALGORITHM	MARCH 2014
---	--	-------------------

Color of Cover: Maroon

(Sample Cover Page and Title Page)

TRANSFER SORT: A PROPOSED SORTING ALGORITHM

KRANZ NIKOLAI V. FONTANILLA

JONALYN N. CORTEZ

RAMIL L. MANANGAN

RHODA T. RONQUILLO

SHYRLEEN M. VALDEZ

**DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
SOUTH LA UNION CAMPUS
COLLEGE OF COMPUTER SCIENCE
AGOO, LA UNION**

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

OCTOBER 2014

The Approval Sheet and the Thesis Committee

- 1) The Approval Sheet contains the title of the study, the researchers, the program graduated, and the date of examination. The Thesis Committee for undergraduate thesis is composed of the following:

Dean
Department Chair
Faculty Research Adviser
Technical Critic
Expert Evaluator

- 2) The Thesis Committee shall be responsible for the conduct and evaluation of the thesis proposal and final examination.
- 3) A faculty is allowed a maximum of ten (10) memberships in the Thesis Committee upon the approval of the Dean and the Department Chair.
- 4) The Chair of the Thesis Committee shall be determined in a collegial basis.
- 5) An Expert Evaluator is an authority in the field of specialization in terms of experience or knowledge. He may be a faculty or a holder of a non-teaching position in the University or from other agencies, public or private.
- 6) The Technical Critic may be specialist in the area of study.
- 7) Faculty Research Adviser shall be chosen on the basis of their expertise in the area of the research study and of their availability.
- 8) The Chair is written first (flush left) followed by the adviser parallel to the right. The other two members are presented in alphabetical order following the left-to-right reading orientation.
- 9) "Chair" shall be used to refer to the Chairperson of the Thesis Committee.
- 10) "Department Chair" shall be used to refer to the Program Chair or Chairperson of the Program.
- 11) The signatories in the undergraduate include the Program Chair and the Dean. The date of signing the manuscript is included.

(Sample Approval Sheet)

APPROVAL SHEET

This thesis, **Transfer Sort: A Proposed Sorting Algorithm**, prepared and submitted by **Kranz Nikolai V. Fontanilla, Jonalyn N. Cortez, Ramil L. Manangan, Rhoda T. Ronquillo and Shyrleen M. Valdez**, in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science was examined and passed on October 2014 by the Thesis Committee composed of:

FLORIBETH P. CUISON, DIT
Chair

EMELY A. MUNAR, MIT
Adviser

ENRICO G. DACANAY, MIT
Member

BELINDA M. DUNGAN, MIT
Member

CHARLIE S. MARZAN, MIT
Member

Accepted and approved in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science.

ENRICO G. DACANAY, MIT
Chairperson, BS Computer Science – Straight Program

Date Signed

FLORIBETH P. CUISON, DIT
Dean, College of Computer Science

Date Signed

Acknowledgments

- 1) The acknowledgment section shall be presented in a formal language, narrative in form, and using the third person point of view. There shall be no pictures or graphics displayed on the page.
- 2) Initials of author(s) with **no** punctuation mark shall be placed at the end of the section, flush to the right.
- 3) There shall be **no** limit as to the number of pages.
- 4) Do **not** use boldface fonts in the body.

Dedication

- 1) This is an optional part of the preliminary.
- 2) This must be brief, centered on one page, with **no** heading.

Table of Contents

Parts

There shall be five (5) main parts written in upper case:

- a. PRELIMINARIES
- b. CHAPTERS
- c. LITERATURE CITED
- d. APPENDICES
- e. CURRICULUM VITAE

The Preliminaries

- a. The word "Preliminaries" shall **not** be written explicitly on the page. The first entry is TITLE PAGE.
- b. The item "Table of Contents" shall **not** be included in the Table of Contents.

The Chapters

- a. The word CHAPTER (in upper case) follows the preliminary entries.
- b. Chapter Titles are indented (0.5 TAB) and written in UPPER CASE.
- c. Items in every chapter are indented (twice 0.5 TAB) and written in Title Case .
- d. Chapter Title shall be numbered using the Hindu-Arabic numeral.

Page Numbers

- a. The word "Page" appears in the Table of Contents.
- b. Set the page numbers in the Table of Contents flush to the right margin.
- c. Dots shall be used as leaders, with **no** space between dots (.....)
- d. *Pagination in the manuscript:* The page numbers for the preliminary parts shall use the lower case Roman Numeral (i, ii, iv, ix, etc.) appearing as footnote, and following a center alignment. The page numbers for the main parts and end matters shall be continuous, using the Hindu-Arabic numeral. However, the pagination at the start of the each shall be hidden. (Tip: superimposes the page number with a borderless blank box).

Secondary Contents

The List of Tables, List of Figures, and List of Plates shall only reflect those that appear in the text.

Tables, Figures, Plates which are in support for basic/primary data in the text shall be placed in the appendices.

List of Tables

- a. Only the tables that appear in the text shall be included in the list.
- b. Numbering (in Hindu-Arabic form) of tables in the text shall be continuous.
- c. Table title is written in Title Case.

List of Figures

- a. Only the figures (graphs, charts, and other illustrative materials) that appear in the text shall be included.
- b. Numbering (in Hindu-Arabic form) of figures in the text shall be continuous.
- c. Figure title is written in Title Case.

List of Plates

- a. Only the plates (pictures) that appear in the text shall be included.
- b. Screen Prints shall be included in the List of Plates.
- c. Numbering (in Hindu-Arabic form) of plates in the text shall be continuous.
- d. Plate title is written in Title Case.

(Sample Acknowledgments)

ACKNOWLEDGMENTS

...no speaker is the first one who disturbs the eternal silence of the universe...

- Bakhtin

...and no book or research can ever be written without reference to those who keep disturbing the universe; hence, this token of appreciation to the following people who have inconveniently but inspiringly stirred the heart and mind of the researchers:

The researchers wish to express their profound gratitude to the following who in one way or the other have contributed to the success of this study.

To the Almighty God for His eternal love, blessings and guidance, and the wisdom He has given us;

To (mention the persons whom you wish to thank).

INITIALS OF AUTHORS

(Sample Table of Contents)

TABLE OF CONTENTS

	Page
TITLE PAGE	i
APPROVAL SHEET	ii
ACKNOWLEDGMENTS	iii
DEDICATION	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
LIST OF PLATES	x
ABSTRACT	xi

CHAPTER

1 INTRODUCTION

Situation Analysis	1
Statement of Objectives	7
Time and Place of the Study	9
Definition of Terms	9

2 METHODOLOGY

Research Design	11
Materials and Procedures	12
Data Gathered	18

	Data Analysis	24
3	RESULTS AND DISCUSSIONS	
	Description of the Existing Algorithms	27
	Description of the Proposed Algorithm	30
	Degree of Validity of the Proposed Algorithm	33
4	SUMMARY, CONCLUSIONS AND RECOMMENDATIONS	
	Summary	38
	Conclusions.....	39
	Recommendations.....	40
	LITERATURE CITED	41
	APPENDICES	
A	Source Code/ Sample Programs of Sorting Algorithms	43
B	Time complexity	44
C	Content Validation for Test/ Evaluation Instruments	45
	CURRICULUM VITAE	
	Researcher's name	53
	Researcher's name	55

LIST OF TABLES

Table No		Page
1	Common Complexity Cases.....	13
2	Results of the Degree of Validity of the Questionnaire in Validating an Algorithm.....	15
3	Comparison of Algorithms Using Reversed Samples	25

LIST OF FIGURES

Figure No		Page
1	Research Paradigm.....	5

LIST OF PLATES

Plate No		Page
1	Bubble Sort Graph.....	16
2	Radix Sort Graph.....	18
3	Comparison of Algorithms Using Reversed Samples.....	25
4	Comparison of Algorithms Using Random Samples (up to 2 digits)	27
5	Comparison of Algorithms Using Random Samples (up to 3 digits).....	28
6	Comparison of Algorithms Using Random Samples (up to 4 digits)	29

The Abstract

- 1) The Abstract is an abridged version (maximum of 150 words) of the most important results of the study.
- 2) It is a one-page part of the preliminary which includes the following:
 - a. one sentence situation//background of the study
 - b. the objectives/problems of the study
 - c. the methodology and
 - d. the salient findings
- 3) Conclusions and recommendations shall **not** be included in the abstract.
- 4) The Abstract shall be presented as follows (note the parts in bold face):

NAME. (Year). Title. Degree (fully spelled out). School, College, Address.
- 5) The name of the Adviser shall be in bold Title Case.
- 6) The header ***Keywords*** (in italics) shall be placed below the abstract. The key words immediately follow after the colon separator and listed in alphabetical order.
- 7) The maximum number of key words is five (5). Pair words/phrases are counted individually.
- 8) The body of the Abstract shall be in *italics*, single spaced.

(Sample Abstract 1)

ABSTRACT

FONTANILLA, KRANZ NIKOLAI V., CORTEZ, JONALYN N., MANANGAN, RAMIL L., RONQUILLO, RHODA T., AND VALDEZ, SHYRLEEN M. (2014). Transfer Sort: A Proposed Sorting Algorithm. Bachelor of Science in Computer Science. Don Mariano Marcos Memorial State University South La Union Campus, College of Computer Science, Agoo, La Union.

Adviser: Munar, Emely A., MIT

Algorithms have become an aid in deciphering various problems.

This study aimed at proposing a sorting algorithm. Specifically, it described existing sorting algorithms in their best, worst, and average cases, which became the basis of the proposed algorithm. The researchers used descriptive and developmental research methods.

The proposed algorithm, Transfer Sort, has this characteristic of transferring elements to a temporary array. It has a best case of $O(n)$, worst case of $O(n^2)$, and average case of $O(n \log n)$. Transfer sort was found to be faster than Bubble sort and Selection sort in all cases; slower than Quick sort but more stable, and slower than Radix sort when $k < \log n$; faster than Insertion sort when sorting reversed samples; and was found to be very highly valid as revealed in the resulting mean of the conducted validity evaluation.

Keywords: average case, best case, Big O notation, sorting algorithm, worst case

(Sample Abstract 2)

ABSTRACT

BOADO, TRINMAR A., CELI, SONNY R., GARCIA, MARIVIC G., MABANGLO, SARAH F., MACALABA, MOHAMMAD ARVIN JAY M., AND NARVASA, DAN RUSTY Q. (2014). Small Enterprise Technology Upgrading Program Web-Based Project Monitoring System of Department of Science and Technology Regional Office 1. Bachelor of Science in Computer Science. Don Mariano Marcos Memorial State University South La Union Campus, College of Computer Science, Agoo, La Union.

Adviser: Marzan, Charlie S., MIT

Web-based applications intensively transforms and boost project monitoring system in all types of agencies.

This study aimed at developing a Small Enterprise Technology Upgrading Program – Web-Based Project Monitoring System of Department of Science and Technology Regional Office 1 (SETUP-WPMS DOST-RO1) to make project monitoring system efficient and effective. Descriptive and developmental research methods were used to organize the presentation, prescription and interpretation of the data gathered.

After the analytical discussions, the following findings were yielded: the assessment of the monitoring of projects was moderately complete, accurate, reliable, timely and secure; SETUP-WPMS DOST-RO1 was developed using RAD model; and SETUP-WPMS was assessed highly valid, highly complied and very usable.

Keywords: *compliance, SETUP-WPMS DOST-RO1, usability, validity*

Part 3

THE MAIN BODY OF THE RESEARCH

Chapter 1

INTRODUCTION

Situation Analysis

The Situation Analysis presents the background of the problem. It is a discussion of the *what's* and *why's* of the study – why the problem was selected and what are the situations under which the problem exists. The analysis should highlight questions, issues, problems, and disturbing observations as well as identify a gap in the field of knowledge which the research intends to address.

Framework of the Study

The framework is a presentation of major theories and/or concepts from which the study was derived and anchored. It is drawn from the review of literature and studies which shapes the research problem.

The *Framework of the Study* presents specific and well-defined constructs, assumptions, expectations, and beliefs that support the study. It uses theories, abstract concepts, facts, laws, variables, and their relations that explain and predict how an observed phenomenon exists and operates. It should link the study to theories/concepts that are useful for interpreting, criticizing, and unifying established scientific laws or facts that serve as guide in discovering new generalizations. In the paper, be explicit as to whether the theory will be verified or developed.

In writing this part, first enumerate the theories/concepts. Prefatory or preview statement before detailed discussion prepares and guides readers of the long discussion that follows. Preferably, present the theories in a diagram form showing their interrelationship and their relation to the current study. The diagram serves as a "framework at a glance" which provides the readers a bird's eye view or general perspective of all the theories and concepts presented. Presenting the theories in a diagram also shows your ability to isolate and analyze a multitude of information, extracting only those which are relevant to your study. At the same time, your discussion of the framework should show your ability to synthesize all these theories and assemble them in a manner that will make your readers see clearly the foundation of your study.

The end part of the discussion of the Framework of the Study should lead to the **research paradigm** (take note that this is **not** written as a heading). The discussion of theories/concepts should naturally lead to the paradigm of the study. It is an offshoot of the theoretical/conceptual framework discussed and must comprehensively show the interrelationship of the variables under study and the direction of the research.

Statement of the Problem/Statement of Objectives

This part is composed of two (2) sections: the general problem/objective and the specific problems/objectives.

The general problem/ objective is cascaded from the title and stated in declarative form. The specific problem/objective section breaks down the general problem/objective into manageable units.

The specific problems are stated in interrogative form and concretize the construct stated in the general problem.

The specific problems/ objectives must be arranged logically from the first step to the last step of the research process to be undertaken. This is to say that when all the specific problems/ objectives are answered/ achieved, the general problem/ objective will have been answered as well.

For Technical Research, the specific objectives are stated in declarative form (i.e., using the *to + verb* form).

Definition of Terms

This section contains terms that are relevant in understanding the study. Definition may be **conceptual** (i.e. taken from books, dictionary, etc) and presented verbatim if the term, as used in the study, has the same definition as written in the cited source/s. However, when the term has a peculiar use in the study or the term requires more specific definition, then an **operational** definition (i.e., how a term is specifically used in the study) be used.

The terms to be defined for this section must be extracted from the *title*, *paradigm of the study*, and *statement of problems/ objectives*. Technical terms appearing in other sections of the manuscript are excluded from this section. They are best defined in the section where they appear.

For presentation, the terms defined in this section are indented, italicized, and listed in alphabetical order. The term and the definition must be presented in sentence form, that is, no period or hyphen should be used to separate them.

Example:

For a better understanding of this research report, the following terms are conceptually and operationally defined:

Algorithms refer to the processes followed in order to solve a certain problem.

Asymptotic behaviour refers to growth of time of an algorithm as the number of inputs approaches infinity.

The Review of Literature

The Review of Literature (ROL) used to be a separate chapter in the old format. In the new DMMMSU format, the ROL is no longer part of the Capstone, Thesis or Dissertation. This does **not** mean, however, that literature review is no longer needed in the manuscript.

In the DMMMSU format, the literature review must be seamlessly incorporated in the *Situation Analysis* (for findings indicating problems that need to be addressed), *Framework of the Study* (for researches/concepts that strengthen or falsify a theory), *Methodology* (for citations on the use of research designs, instruments/ materials and statistical tools used) and in the *Results and Discussion* section (pointing out similarities and differences of previous studies and the

current findings and citing theories that explain a particular finding especially the unexpected results).

Below shows the introductory parts of a technical and social science researches respectively:

TECHNICAL RESEARCH	SOCIAL RESEARCH
• Situation Analysis (Side Heading) Discussion on the what's and why's and significance/importance of the study. • Statement of Objectives Both general and specific objectives shall be stated in declarative form. • Time and Place of the Study • Definition of Terms	• Situation Analysis (Side Heading) Discussion on the what's and why's of the study. Embedded in the discussion is the significance/importance of the research that justifies the study • Framework of the Study (with Paradigm) Presentation of major theories/ concepts from which the study is anchored. Introduce the theories/concepts in diagram form showing their interrelationship and their relation to the current study. The discussion of theories/concepts should lead to the paradigm of the study which must comprehensively explain the relationship of the variables and the direction of the study. • Statement of the Problem General/main problem shall be in declarative form and the specific problems shall be in interrogative form. • Definition of Terms

Chapter 2

METHODOLOGY

The methodology part presents the design, materials and procedures, data gathered, and data analysis to accomplish the objectives.

Research Design

This section specifies the general method and specific design used. The description of the design must include an explanation of the selected design and a justification of its appropriateness in relation to the objective/s of the research and data to be gathered.

Some of the research designs include the Exploratory Research, Descriptive Research, Causal Research, and Developmental Research.

Exploratory research is used to discover, formulate problems more precisely, develop hypotheses, establish priorities for research, eliminate impractical ideas, and clarify concepts. Types of exploratory research include: literature search, experience survey, analysis of select cases, interviews, ethnographies (developing understandings of the everyday activities of people in local settings), and focus groups (small group discussions led by a trained moderator) to generate ideas, understand consumer vocabulary, reveal consumer needs, motives, perceptions, and attitudes on products and services.

Descriptive research involves asking a sample of people from a population a set of questions and using the answers to *describe* that population (survey research). This type is used to describe what is going on or exists, estimate how groups of consumers might behave, and examine relationships between two or more variables. It is used to describe segment characteristics, estimate proportion of people who behave in a certain way, and make specific predictions. Its types include: Longitudinal study, Panels, and Sample Survey

Causal Research helps determine if one or more IVs (independent variables) (treatment, predictors, or causes) affects one or more DVs (outcome variables). This design requires the highest degree of understanding of the problem. It is used to provide evidence regarding causal relationships and rule out all other explanations. One type of causal research is the Experimental Research that involves scientific investigation in which an investigator manipulates and controls one or more independent variables and observes the dependent variable for variation concomitant to the manipulation of the independent variables. This includes Laboratory Experiment in which investigator creates a situation with exact conditions, so as to control some, and manipulate other, variables; and Field Experiment which is a research study in a realistic situation in which one or more independent variables are manipulated by the experimenter under as carefully controlled conditions as the situation will permit. Experimental designs may be One Group Pretest-Posttest Design or Pretest Post-Test Control Group Design.

Developmental research is "the systematic study of designing, developing and evaluating instructional programs, processes and products that must meet the criteria of internal consistency and effectiveness" (Seels & Richey, 1994, p. 127). It is a pragmatic type of research that offers a way to test "theory" that has been only hypothesized and to validate practice that has been perpetuated essentially through unchallenged tradition. In addition, it is a way to establish new procedures, techniques, and tools based upon a methodical analysis of specific cases. There are two categories of developmental research, referred to as Type 1 and Type 2 (Richey, Klein, & Nelson, 2004). Type 1 developmental studies focus upon a given instructional product, program,

process, or tool. They reflect an interest in identifying either general development principles or situation-specific recommendations. Typically Type 1 studies address not only product design and development, but evaluation as well. At times they may validate a particular design or development technique or tool. Type 2 studies, on the other hand, focus upon a given design, development, or evaluation model or process. They may involve constructing and validating unique design models and processes, as well as identifying those conditions that facilitate their successful use. (Richey, 2005).

Materials and Procedures (For Technical Researches)

It lists down the materials used (questionnaires, tests, interview schedule, observation checklist, or mechanical gadgets, etc). It details the processes conducted in adoption, modification, or development of the instrument as well as the procedures conducted to ensure the validity and reliability of the instrument. It shall also detail the procedures undertaken in the conduct of the experiment/study. The procedure may be shown in diagram form and explained afterwards.

The population/ respondents of the study are also mentioned in this section. The procedure on obtaining the sample respondents (where necessary) shall be explained.

Data Gathered (For Technical Researches)

This presents/ describes the data to be gathered for each objective.

Sources of Data (For Social Researches)

This section identifies the population of the study and samples where necessary. It includes the sampling method/formula/data used, and justification for the choice of the sampling method/formula. Description and profiling of respondents/subjects in terms of number and demographic characteristics are presented in this section except when such is part of the statement of the problem and thus needs to be presented in the Results and Discussion section.

Also included are the locale, and time of the study.

Instrumentation and Data Collection (For Social Researches)

The first section presents the instruments used (questionnaires, tests, interview schedule, observation checklist, or mechanical gadgets, etc) and the parts. It details the processes conducted in adoption, modification, or development of the instrument as well as the procedures conducted to ensure the validity and reliability of the instrument. An explanation is necessary to justify the choice of the data collection tool/s. As much as possible, the research should aim for triangulation (use of several data-gathering tools) to strengthen the validity of results.

The second section presents the data to be gathered and detailed description of the means of gathering the said data – techniques and procedures in data gathering as well as the manipulations and controls made such as the assignment of subjects into control and experimental groups.

Analysis of Data/ Data Analysis

This section specifies the statistical tool/s used in the treatment of the data gathered (e.g. frequency count, percentage, mean, weighted mean, t-test, ANOVA, chi-square, Pearson r , etc).

Also included are guides (range values and descriptors) for the categorization and interpretation of data, including levels of significance/correlation needed in testing the hypothesis, and software (with corresponding version) used. If unfamiliar statistical tests are used, the formula is explained.

Below is a summary of the parts of Chapter 2 for both Technical and Social Science Researches:

TECHNICAL RESEARCH	SOCIAL RESEARCH
• Research Design • Materials and Procedures • Data Gathered • Data Analysis	• Research Design • Sources of Data • Instrumentation and Data Collection • Analysis of Data

Note: For research involving system development, system features or output must be agreed upon by the client, the researchers and the Thesis Committee members during the system proposal, and must be documented using a Compliance Matrix. Further any recommendations made during the proposal shall be documented using the appropriate Compliance Matrix.



Don Mariano Marcos Memorial State University
South La Union Campus
COLLEGE OF COMPUTER SCIENCE
Agoo, La Union



COMPLIANCE MATRIX
(System Presentation)

Title of Study
Proponents/ Researchers
Name of Client/ Company
Date

.....
.....
.....
.....

Instructions: Please put a check mark (/) on the space provided to indicate whether the feature has been complied or not complied.

System Features/ Expected System Output	Complied	Not Complied
1)		
2)		
3)		
:		
:		
:		
n)		

This is to certify that the aforementioned features were recommended and agreed upon during the system proposal, and that all features expected of the system must be complied before it can finally be accepted by the undersigned representing the client.

Client (Signature over printed name)

Noted:

Thesis Committee

Adviser

Member

Member

Member

Chair



Don Mariano Marcos Memorial State University
South La Union Campus
COLLEGE OF COMPUTER SCIENCE
Agoo, La Union



COMPLIANCE MATRIX
(System Proposal)

Title of Study
Proponents/ Researchers
Name of Client/ Company
Date

	Expected System Features/ Output
1)	
2)	
3)	
:	
:	
:	
n)	

This is to certify that the aforementioned system features/ outputs were recommended and agreed upon by the client, the researchers and the Thesis Committee members, and are expected to be exhibited/ performed by the system during the system presentation.

Adviser

Member

Client

Member

Member

Student Leader

Thesis Committee Chair

NOTE: System features or output must be agreed upon by the client, the researchers and the Thesis Committee members during the system proposal, and must be documented using the Compliance Matrix (System Proposal).

Student Researchers are required to accomplish the Compliance Matrix form (System Presentation) with the client. The signed form must be presented during the Final Thesis defense.

Chapter 3

RESULTS AND DISCUSSION

This section involves presentation, analysis, and interpretation of data.

Data are broken down into component parts and presented in the proper order and those that are pertinent and relevant and reducing them to form that are intelligible and interpretable so that the relationships between the research problem and their intended answers can be studied. Research data that have been subjected to analysis can be presented as text or using tables, graphs, and other pictorial forms.

Presented data must also be interpreted. This is done by making inferences relevant to the research questions and from where generalizations are drawn. Key in interpretation is finding meanings and drawing implications.

Interpretation is done while the results are being presented and analyzed, then drawing relationships and linking to existing theories and findings.

Topic Headings

The discussion of results shall be guided by the statement of problems or objectives. The number of main topic headings should be parallel to the number of problems or objectives.

Presentation of Findings

Findings may be presented in textual, tabular, graphical/pictorial form or a combination of these three depending on the appropriateness for such.

In answering every problem statement/objective of the study, the following parts and guidelines should be observed:

a. *Prefatory Information.*

Make a one paragraph introduction of the problem.

b. *Introduce table/figure/chart*

Make a one-sentence introduction describing the content of the table/graph/plate presented.

c. *Where possible, the table/ figure/ chart/ plate must be displayed within the same page after the introduction. If it cannot be displayed in a single page due to its size, it can be transferred to the immediate page following the presentation but a signal phrase must be included referring to it (e.g. Table 5 on page 87...).*

d. *The table/ figure/ chart/ plate is followed with a textual discussion of the data presented. In case of the table/figure/chart/plate transferred to the following page, the textual discussion should be advanced in the previous page to avoid creating a white space/gap within a chapter).*

e. In the discussion part, present only the critical data/results. Let the table speak for itself. **DO NOT STATE WHAT IS OBVIOUS IN THE TABLE.** If data are numerous and repetitive, reading becomes boring; hence, **only highlights or important parts of the data are textually presented.**

f. In reading/explaining a table, observe the following:

i. Use a logical form of presentation

Present a general observation (e.g., the grand mean or average mean followed by specific observations (weighted means by cluster) or vice versa.

ii. Extract the significant findings (e.g. top three and last three findings and the average)

g. Descriptive equivalents/interpretation should be "seamlessly embedded" in the discussion and **not** presented as it appears in the table.

Wrong: The table shows that Indicator 1 which states "The leader is democratic...." has an average weighted mean of 4.50 which means "very much evident" as rated by the respondents.

Correct: The table shows that democratic process is **very much evident** in the institution based on the computed average weighted mean of 4.50.

Correct: It can be glossed from the table that democratic processes are observed in the institution and **very much evident** (AWM=4.50) in the system as perceived by the three groups of respondents.

h. Textual presentation must be followed by **interpretation, inference, or implication.** The following elements are expected in the discussion:

i. Statement of Results

Explain why the results came out as such. If results are unexpected, be able to provide an explanation.

In making interpretations, observe the following elements:

- Condition contributing to the result
- Cause that brings about the result
- Effect or impact of the result
- Suggestions related to the result

ii. Statement of Hypothesis (when this is expressed in the Statement of the Problem/ Objectives)

Take note that the findings and claim **come after** the statement of hypotheses (and acceptance/rejection of such).

Wrong: However, when subjected to the t-test analysis, results revealed **no significant difference** as indicated by the p-value of 1.093 which is

lower than the tabulated p-value of 2.228 at 0.05 level of significance. Therefore, the null hypothesis which states that there is no significant difference between the perceptions of the expert evaluators and the teacher-respondents as the extent of suitability of the xxx is accepted.

Wrong: When subjected to t-test analysis, results revealed no significant difference as indicated by the p-value of 1.093 which is lower than the tabulated p-value of 2.228 at 0.05 level of significance. Therefore, the null hypothesis which states that... is accepted.

Correct: When subjected to t-test analysis, the p-value of 1.093 is lower than the tabulated p-value of 2.228 at 0.05 level of significance. Therefore, the null hypothesis which states that... is accepted. This result reveals that there is no significant difference between the ...

- iii. Compare results with previous findings/theories to make generalizations
- iv. Draw conclusion/ make a claim
- v. Explain the implication of the finding/conclusion to the field of study or to concerned/affected groups
- vi. Include recommendations from other researches (as needed) as an input/ basis for the current researcher in making a recommendation in the succeeding chapter

Presentation of Table

- a. Table numbering must use continuous coding in Hindu-Arabic numeral (Table 1, 2, 3, etc.). Alphanumeric or decimal coding is **not** allowed.
- b. The table title must be written in Title Case flush to the left margin. In case of a two-liner title, the first letter in the second line should be aligned to the first letter in the first line (parallel indentation).
- c. Table title must **not** be written in boldface or italics.
- d. Table header must **not** be written in boldface or italics.
- e. The table shall be drawn from margin to margin.
- f. Use a compound line (same thickness) for the top and bottom lines of the table (Top and Bottom line weight is 1 ½ pt ; Inside line weight is 1 pt).
- g. Use a single line to separate the grand mean.
- h. Use single space if there are more than 10 entries in the table.
- i. Use double space if there are less than 10 entries in a table.
- j. The MEAN shall be entered up to two decimal places only. Also, for uniformity of presentation, use two decimal places (e.g. 9.50 not 9.5).
- k. Properly align numerical values when presenting data (ones, tens, hundreds, etc.). Add zero before the decimal point for value less than 1.

98.50
112.25
0.85
- l. The font size may be reduced up to font size 10 (Times New Roman).
- m. Single space separates the table title and table.
- n. Double space separates the table header and the first entry line.

- o. Use the symbol **M** to refer to Mean and **WM** to refer to Weighted Mean.
- p. Use the symbol (%) in the table and when enclosed in parentheses; use the word *percent* in the textual discussion part.
- q. As much as possible, tables/figures/plates shall not be placed (or sandwiched) between paragraphs.
- r. In case of long tables and there is a need to cut the table, the first part should always be longer than the second part. For the second table, add **Continuation of Table <no.>** as header. Cut tables properly; data that go together must not be separated. Include header in the second .
- s. If the remaining part of a cut table consists of 2-3 lines only, the font size may be reduced (up to font 10) so that the table can be accommodated in just one page
- t. Indicate in the legend the necessary interpretation of symbols used in the table. Include also the whole range values, descriptive phrases, and acronyms used in categorizing and interpreting data. Font 10 shall be used for the *Legend*.
- u. Use asterisk (*) to denote significant finding. No column should be used to indicate acceptance/rejection of hypotheses and significance/insignificance of findings.

Example

text text text text text text text text text text text text text text text text text text text
(2 single spaces separate the text and table title)

Table 1. Title of Table is Written in Title Case and Not Printed in Boldface/Italics

xx

(1 single space separates the table title and table)

Table Heading is not written in boldface; allow some breathing space inside the table

Top and Bottom	Inside	line
Line weight is	weight is	1 pt
1.5 pts		

(Single space separate the last line of the table and text)

Legend:

text text text text text text text text text text text text text text text text text text

(Sample Table)

Table 2. Number of Phases Using Reversed Samples

No. of Elements	Bubble Sort	Selection Sort	Radix Sort	Insertion Sort	Quick Sort	Transfer Sort
10	81	54	20	45	14	19
20	361	209	40	190	29	39
30	841	464	60	435	44	59
40	1521	819	80	780	59	79
50	2401	1274	100	1225	74	99
60	3481	1829	120	1770	89	119
70	4761	2484	140	2415	104	139
80	6241	3239	160	3160	119	159
90	7921	4094	180	4005	134	179
100	9801	5049	300	4950	149	199
200	39601	20099	600	19900	299	399
300	89401	45149	900	44850	449	599
400	159201	80199	1200	79800	599	799
500	249001	125249	1500	124750	749	999

(Sample of a Cut Table)

Table 3. Number of Phases Using Random Samples Up to 2 Digits

No. of Elements	Bubble Sort	Selection Sort	Radix Sort	Insertion Sort	Quick Sort	Transfer Sort
10	81	54	20	45	14	19
20	361	209	40	190	29	39
30	841	464	60	435	44	59
40	1521	819	80	780	59	79
50	2401	1274	100	1225	74	99
60	3481	1829	120	1770	89	119
70	4761	2484	140	2415	104	139
80	6241	3239	160	3160	119	159

Continuation of Table3

No. of Elements	Bubble Sort	Selection Sort	Radix Sort	Insertion Sort	Quick Sort	Transfer Sort
90	7921	4094	180	4005	134	179
100	9801	5049	300	4950	149	199
200	39601	20099	600	19900	299	399
300	89401	45149	900	44850	449	599
400	159201	80199	1200	79800	599	799
500	249001	125249	1500	124750	749	999

Presentation of Figure

- Figures shall be enclosed in a box, drawn from margin to margin.
- Needed data, such as percentages/frequencies shall be evident.
- Number and title shall be placed below the enclosed figure.
- Figure numbering must use continuous coding in Hindu-Arabic numeral.
- The figure title must be written in Title Case, center alignment.
- Figure title must **not** be written in boldface or italics.
- Use the abbreviation for the figure label (e.g. Fig. 5)

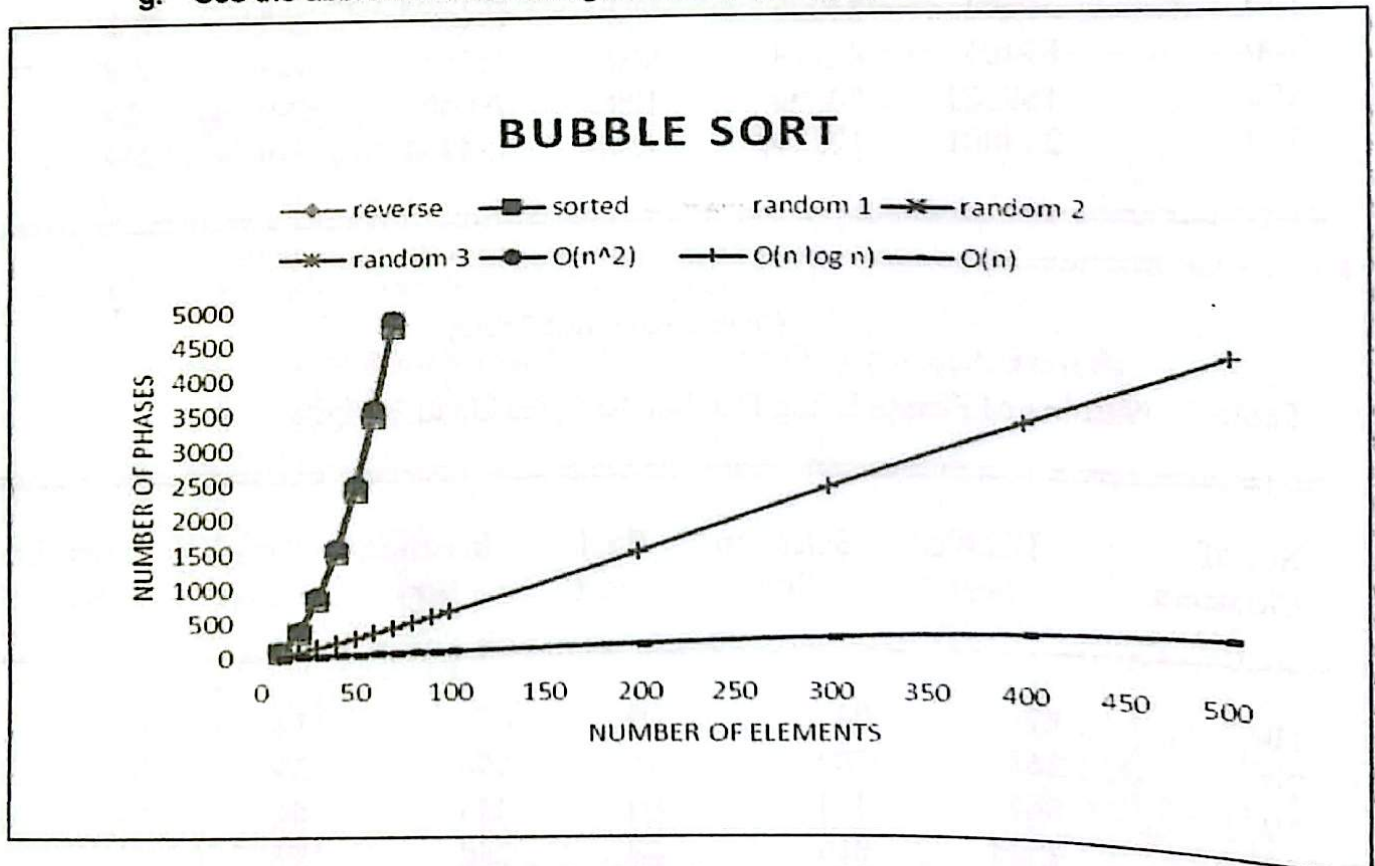


Fig. 1. Title of the Figure Using Title Case (Not Printed in Boldface/Italics)

Presentation of Plate/Photograph

- The size of the plate is preferably half page of the paper.
- The plate shall be drawn from margin to margin.
- Two (2) plates in one page may be allowed.
- Number and title shall be placed below the enclosed plate.
- Plate numbering must use continuous coding in Hindu-Arabic numeral.
- The plate title must be written in Title Case, center alignment.
- Plate title must not be written in boldface or italics.

Year	Title	Refund Rate	Action
2012	Process Improvement, Label and Packaging Upgrading for Exotic Food Products	0.12%	[Icon] [Icon]
2012	Antique Furniture Reproduction Technology Upgrading	0.00%	[Icon] [Icon]
2012	Metallurgy Technology Upgrading	0.00%	[Icon] [Icon]
2012	Upgrading of Antique and Furniture	0.00%	[Icon] [Icon]
2012	Wood Processing Technology Upgrading	0.00%	[Icon] [Icon]
2011	Packaging Improvement Upgrading Technology	0.00%	[Icon] [Icon]
2014	Quality Development and Computer Hardware Upgrades	0.00%	[Icon] [Icon]
2012	Process Enhance for an Auto Service Fabrication Shop	0.00%	[Icon] [Icon]

Copyright © 2013 DOST-RO1. All rights reserved. Developed by trinity01 Systems

Plate 1. Title of the Plate Using Title Case (Not Printed in Boldface/Italics)

Note: For researches involving system development, student researchers are required to accomplish the Compliance Matrix form (refer to pp. 35-37) with the client. The signed form must be presented during the Final Thesis defense.

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary

This part summarizes the entire study. It is composed of an introductory statement about the general problem/objective of the study, the specific problems/objectives, and the methodology employed - all written in paragraph form. This is immediately followed by the salient findings (numbered list). Take note that the "findings of the study" is **not** a separate heading.

Numerical data are included in the "major" findings to be presented. However, no new findings should be presented in this section that have-not been presented in Chapter 3. The number of findings to be presented should be parallel to the number of specific problems/objectives.

Conclusions

The Conclusion Section draws generalization for the population and circumstances for which the evidences have been collected.

It is composed of an introductory statement followed by the conclusions in relation to the findings re: the specific problems/objectives.

The number of conclusions is parallel to the number of specific findings based on the specific problems/objectives. These are presented in numbered list form. The conclusions should not repeat the same words/statement used in the findings section.

No numerical/statistical presentations are made in this section.

Recommendations

This section offers recommendations based on what the results mean in terms of existing knowledge, drawing implications for the field or academic discipline represented by the problem, or the need for further research and policy formulation.

The Recommendation Section is composed of an introductory statement followed by the recommendations in numbered list form corresponding to the conclusions presented. More recommendations may be added where necessary and applicable.

The recommendations should be stated in realistic and practicable (doable) terms identifying individuals or groups who shall carry out the action/s recommended.

Part 4

THE END MATTERS

LITERATURE CITED

This section allows the reader to observe the scope of the investigation behind the report. It also permits the reader to easily locate the full bibliographic information of the materials cited in the text and retrieve any source cited in the body of the paper. Each source cited in the paper must appear in your reference list; likewise, each entry in the reference list must appear in the main body.

Guidelines

- a. The list of reference materials cited in the study is presented as one group with no categorization. These are alphabetically arranged
- b. Electronic sources shall include the date of retrieval.
Author. (Month, Year). Title of article, *Journal Publication*, Volume (Issue), page number/s. Retrieved on (date) from <http://www.literaturecited.com>
- c. Citations/reviews taken from Wikipedia/Blogs are **not** acceptable.
- d. For gray areas **not** specified in this style guide, the APA format (6th edition) must be used.
- e. Secondary citation shall be limited to two (2) only.
- f. References shall be 50% published materials and 50% unpublished materials.
- g. Researches as reference shall be within the last five (5) years; no restriction as to the date of publication of theories, principles, seminal works, and landmark studies.
- h. The minimum number of literature to be reviewed and cited in the manuscript shall be 20 for the Baccalaureate degrees, 30 for the Master's degrees, and 50 for the Doctorate degrees.
- i. The references must follow double spacing, hanging indentation.

Basic Rules (Source: APA Style from Purdue OWL Online Writing Lab, 2015)

- a. All lines after the first line of each entry in your reference list should be indented one-half inch from the left margin. This is called hanging indentation.
- b. Authors' names are inverted (last name first); give the last name and initials for all authors of a particular work for up to and including seven authors. If the work has more than seven authors, list the first six authors and then use ellipses after the sixth author's name. After the ellipses, list the last author's name of the work.
- c. Reference list entries should be alphabetized by the last name of the first author of each work.
- d. For multiple articles by the same author, or authors listed in the same order, list the entries in chronological order, from earliest to most recent.

- e. Present the journal title in full.
- f. Maintain the punctuation and capitalization that are used by the journal in its title.
 - a. For example: *ReCALL* not *RECALL* or *Knowledge Management Research & Practice* not *Knowledge Management Research and Practice*.
- g. Capitalize all major words in journal titles
- h. When referring to books, chapters, articles, or Web pages, capitalize only the first letter of the first word of a title and subtitle, the first word after a colon or a dash in the title, and proper nouns. Do not capitalize the first letter of the second word in a hyphenated compound word.
- i. Italicize titles of longer works such as books and journals.
- j. Do not italicize, underline, or put quotes around the titles of shorter works such as journal articles or essays in edited collections.

NOTE: For more detailed styles and format, refer to the APA 6th Edition. You may also use the MacGraw Hill APA Bibliomaker software for an easier preparation of the bibliography

APA Reference Style Samples

(Source: Purdue OWL Online Writing Lab. (2015). APA style. Retrieved from <https://owl.english.purdue.edu/owl/section/2/10/>)

Format for Books

Author, A. A. (Year of publication). *Title of work: Capital letter also for subtitle*. Location: Publisher.

Note: For "Location," you should always list the city and the state using the two letter postal abbreviation without periods (New York, NY).

Calfee, R. C., & Valencia, R. R. (1991). *APA guide to preparing manuscripts for journal publication*. Washington, DC: American Psychological Association.

Edited Book, No Author

Duncan, G. J., & Brooks-Gunn, J. (Eds.). (1997). *Consequences of growing up poor*. New York, NY: Russell Sage Foundation.

Edited Book with an Author or Authors

Plath, S. (2000). *The unabridged journals*. K. V. Kukil (Ed.). New York, NY: Anchor.

A Translation

Laplace, P. S. (1951). *A philosophical essay on probabilities*. (F. W. Truscott & F. L. Emory, Trans.). New York, NY: Dover. (Original work published 1814)

Note: When you cite a republished work, like the one above, in your text, it should appear with both dates: Laplace (1814/1951).

Edition Other Than the First

Helfer, M. E., Kempe, R. S., & Krugman, R. D. (1997). *The battered child* (5th ed.). Chicago, IL: University of Chicago Press.

Article or Chapter in an Edited Book

O'Neil, J. M., & Egan, J. (1992). Men's and women's gender role journeys: A metaphor for healing, transition, and transformation. In B. R. Wainrib (Ed.), *Gender issues across the life cycle* (pp. 107-123). New York, NY: Springer.

Multivolume Work

Wiener, P. (Ed.). (1973). *Dictionary of the history of ideas* (Vols. 1-4). New York, NY: Scribner's.

Format for Authors

Single Author

Last name first, followed by author initials.

Berndt, T. J. (2002). Friendship quality and social development. *Current Directions in Psychological Science*, 11, 7-10.

Two Authors

List by their last names and initials. Use the ampersand instead of "and."

Wegener, D. T., & Petty, R. E. (1994). Mood management across affective states: The hedonic contingency hypothesis. *Journal of Personality and Social Psychology*, 66, 1034-1048.

Three to Seven Authors

List by last names and initials; commas separate author names, while the last author name is preceded again by ampersand.

Kernis, M. H., Cornell, D. P., Sun, C. R., Berry, A., Harlow, T., & Bach, J. S. (1993). There's more to self-esteem than whether it is high or low: The importance of stability of self-esteem. *Journal of Personality and Social Psychology*, 65, 1190-1204.

More Than Seven Authors

List by last names and initials; commas separate author names. After the sixth author's name, use an ellipses in place of the author names. Then provide the final author name. There should be no more than seven names.

Miller, F. H., Choi, M. J., Angeli, L. L., Harland, A. A., Stamos, J. A., Thomas, S. T., . . . Rubin, L. H. (2009). Web site usability for the blind and low-vision user. *Technical Communication*, 57, 323-335.

Organization as Author

American Psychological Association. (2003).

Unknown Author

Merriam-Webster's collegiate dictionary (10th ed.). (1993). Springfield, MA: Merriam-Webster.

Two or More Works by the Same Author

Use the author's name for all entries and list the entries by the year (earliest comes first).

Berndt, T. J. (1981).

Berndt, T. J. (1999).

When an author appears both as a sole author and, in another citation, as the first author of a group, list the one-author entries first.

Berndt, T. J. (1999). Friends' influence on students' adjustment to school. *Educational Psychologist*, 34, 15-28.

Berndt, T. J., & Keefe, K. (1995). Friends' influence on adolescents' adjustment to school. *Child Development*, 66, 1312-1329.

References that have the same first author and different second and/or third authors are arranged alphabetically by the last name of the second author, or the last name of the third if the first and second authors are the same.

Wegener, D. T., Kerr, N. L., Fleming, M. A., & Petty, R. E. (2000). Flexible corrections of juror judgments: Implications for jury instructions. *Psychology, Public Policy, and Law*, 6, 629-654.

Wegener, D. T., Petty, R. E., & Klein, D. J. (1994). Effects of mood on high elaboration attitude change: The mediating role of likelihood judgments. *European Journal of Social Psychology*, 24, 25-43.

Format for Periodicals

Article in Journal Paginated by Volume

Harlow, H. F. (1983). Fundamentals for preparing psychology journal articles. *Journal of Comparative and Physiological Psychology*, 55, 893-896.

Article in Journal Paginated by Issue

The parentheses and issue number are not italicized or underlined.

Scruton, R. (1996). The eclipse of listening. *The New Criterion*, 15(3), 5-13.

Article in a Magazine

Henry, W. A., III. (1990, April 9). Making the grade in today's schools. *Time*, 135, 28-31.

Article in a Newspaper

Unlike other periodicals, p. or pp. precedes page numbers for a newspaper reference in APA style. Single pages take p., e.g., p. B2; multiple pages take pp., e.g., pp. B2, B4 or pp. C1, C3-C4.

Schultz, S. (2005, December 28). Calls made to strengthen state energy policies. *The Country Today*, pp. 1A, 2A.

Letter to the Editor

Moller, G. (2002, August). Ripples versus rumbles [Letter to the editor]. *Scientific American*, 287(2), 12.

Review

Baumeister, R. F. (1993). Exposing the self-knowledge myth [Review of the book *The self-knower: A hero under control*, by R. A. Wicklund & M. Eckert]. *Contemporary Psychology*, 38, 466-467.

Format for Other Print Sources

An Entry in an Encyclopedia

Bergmann, P. G. (1993). Relativity. In *The New Encyclopedia Britannica*. (Vol. 26, pp. 501-508). Chicago, IL: Encyclopedia Britannica.

Work Discussed in a Secondary Source

List the source the work was discussed in:

Coltheart, M., Curtis, B., Atkins, P., & Haller, M. (1993). Models of reading aloud: Dual-route and parallel-distributed-processing approaches. *Psychological Review*, 100, 589-608.

NOTE: Give the secondary source in the references list; in the text, name the original work, and give a citation for the secondary source. For example, if Seidenberg and McClelland's work is cited in Coltheart et al. and you did not read the original work, list the Coltheart et al. reference in the References. In the text, use the following citation:

In Seidenberg and McClelland's study (as cited in Coltheart, Curtis, Atkins, & Haller, 1993), ...

Dissertation/Thesis Abstract

Yoshida, Y. (2001). Essays in urban transportation. *Dissertation Abstracts International*, 62, 7741A.

Dissertation/Thesis, Published

Lastname, F. N. (Year). *Title of dissertation*. (Doctoral dissertation). Retrieved on January 10, 2015 from Name of database. (Accession or Order Number)

Dissertation/Thesis, Unpublished

Lastname, F. N. (Year). *Title of dissertation*. (Unpublished doctoral dissertation). Name of Institution, Location.

Government Document

National Institute of Mental Health. (1990). *Clinical training in serious mental illness* (DHHS Publication No. ADM 90-1679). Washington, DC: U.S. Government Printing Office.

Report From a Private Organization

American Psychiatric Association. (2000). *Practice guidelines for the treatment of patients with eating disorders* (2nd ed.). Washington, DC: Author.

Conference Proceedings

Schnase, J. L., & Cunnius, E. L. (Eds.). (1995). *Proceedings from CSCL '95: The First International Conference on Computer Support for Collaborative Learning*. Mahwah, NJ: Erlbaum.

Format for Electronic Sources

Article From an Online Periodical

Bernstein, M. (2002). 10 tips on writing the living Web. *A List Apart: For People Who Make Websites*, 149. Retrieved on January 5, 2015 from <http://www.alistapart.com/articles/writeliving>

Article From an Online Periodical with DOI Assigned

Wooldridge, M.B., & Shapka, J. (2012). Playing with technology: Mother-toddler interaction scores lower during play with electronic toys. *Journal of Applied Developmental Psychology*, 33(5), 211-218. <http://dx.doi.org/10.1016/j.appdev.2012.05.005>

Article From a Database

Smyth, A. M., Parker, A. L., & Pease, D. L. (2002). A study of enjoyment of peas. *Journal of Abnormal Eating*, 8(3), 120-125. Retrieved on January 5, 2015 from <http://www.articlehomepage.com/full/url/>

Abstract

Paterson, P. (2008). How well do young offenders with Asperger Syndrome cope in custody?: Two prison case studies [Abstract]. *British Journal of Learning Disabilities*, 36(1), 54-58. Retrieved on January 5, 2015 from <http://www.articlehomepage.com/full/url/>

Newspaper Article

Parker-Pope, T. (2008, May 6). Psychiatry handbook linked to drug industry. *The New York Times*. Retrieved on January 5, 2015 from <http://well.blogs.nytimes.com>

Dissertation/Thesis from a Database

Biswas, S. (2008). *Dopamine D3 receptor: A neuroprotective treatment target in Parkinson's disease*. Retrieved on January 5, 2015 from ProQuest Digital Dissertations. (AAT 3295214)

Online Encyclopedias and Dictionaries

Feminism. (n.d.). In *Encyclopædia Britannica online*. Retrieved on January 5, 2015 from <http://www.britannica.com/EBchecked/topic/724633/feminism>

Data Sets

United States Department of Housing and Urban Development. (2008). *Indiana income limits* [Data file]. Retrieved on January 5, 2015 from http://www.huduser.org/Datasets/IL/IL08/in_fy2008.pdf

Graphic Data (e.g. Interactive Maps and Other Graphic Representations of Data)

Solar Radiation and Climate Experiment. (2007). [Graph illustration the SORCE Spectral Plot May 8, 2008]. *Solar Spectral Data Access from the SIM, SOLSTICE, and XPS Instruments*. Retrieved on January 5, 2015 from http://lasp.colorado.edu/cgi-bin/ion-p?page=input_data_for_spectra.ion

Qualitative Data and Online Interviews

Butler, C. (Interviewer) & Stevenson, R. (Interviewee). (1999). *Oral History 2* [Interview transcript]. Retrieved on January 5, 2015 from Johnson Space Center Oral Histories Project Web site: http://www11.jsc.nasa.gov/history/oral_histories/oral_histories.htm

Online Lecture Notes and Presentation Slides

Roberts, K. F. (1998). *Federal regulations of chemicals in the environment* [PowerPoint slides]. Retrieved on January 5, 2015 from <http://siri.uvm.edu/ppt/40henv/index.html>

Computer Software/Downloaded Software

Hayes, B., Tesar, B., & Zuraw, K. (2003). OTSoft: Optimality Theory Software (Version 2.1) [Software]. Available from <http://www.linguistics.ucla.edu/people/hayes/otsoft/>

E-mail

E-mails are not included in the list of references, though you parenthetically cite them in your main text: (E. Robbins, personal communication, January 4, 2001).

Online Forum or Discussion Board Posting

Frook, B. D. (1999, July 23). New inventions in the cyberworld of toylandia [Msg 25]. Message posted to <http://groups.earthlink.com/forum/messages/00025.html>

Blog (Weblog) and Video Blog Post

J Dean. (2008, May 7). When the self emerges: Is that me in the mirror? [Web log comment]. Retrieved on January 5, 2015 from <http://www.spring.org.uk/the1sttransport>

Video Podcasts

Scott, D. (Producer). (2007, January 5). The community college classroom [Episode 7]. *Adventures in Education*. Podcast retrieved on January 5, 2015 from <http://www.adveeducation.com>

APPENDICES

The Appendix Section immediately follows the Reference Section. It is introduced by a cover page labeled APPENDICES (displayed in the center of the page, font 16).

The Appendices guarantee a check upon the validity and truthfulness of the research evidences presented in the text and may include the following:

1. Materials, due to length or complex nature, that cannot be readily reproduced in the text (maps, elaborate graphs, computer printouts)
2. Samples of instruments used
2. Illustrative materials such as scores in tests, computation tally sheets
3. Documents pertinent to the problem, otherwise not presentable in length in the text like laws, decrees, supportive legal decisions
4. Operating instruments to support the action plan such as memorandum, program of instruction, policy proposal, syllabus, etc.
5. Use uppercase for the label APPENDIX A and title case for the title (in single space)
6. The **alphabet** should be used for coding; no use of alphanumeric or decimal coding (A-Z; AA-ZZ)

The appendices must be exhibited in continuous sequence following the sequence categories listed below:

TECHNICAL RESEARCH	SOCIAL RESEARCH
A. Tables B. Figures C. Plates D. Others (Questionnaire/ Instrument, Tally of Results, List of Presentations and Publications, Communication, Others)	A. Communication B. Questionnaire/Instrument C. Tables D. Figures E. Plates E. Others (Tally of Results, List of Presentations and Publications, Others)

Example:

APPENDIX A **SAMPLE USABILITY QUESTIONNAIRE** **Website Usability Contingent Evaluation Tool** **Adopted from Dunstan (2003)**

Name: _____
Position: _____

Date: _____

Direction: Please put a check mark (/) under the rating that best describes the indicators using the following rating scales to determine the level of usability of the developed system.

Scale:

- 5 - Strongly Agree
- 4 - Agree
- 3 - Moderately Agree
- 2 - Disagree
- 1 - Strongly Disagree

Usability Dimension	Questions	5	4	3	2	1
a. Content	1. The content of this web site is useful for its intended purpose and audience.					
	2. This web site is accurate or truthful with the information presented.					
	3. Information presented at this site is complete.					
b. Clarity/ Presentation	1. This web site is easily readable and relatively easy to understand.					
	2. The layout of this web site lends itself to good planning of activities.					
	3. This web site is not cluttered with unnecessary diagrams/graphics/etc.					
c. Hyperlinks	1. The hyperlinks on this web site are adequate.					
	2. The works used to describe the hyperlink (i.e. underlined) are clear.					
	3. The hyperlinks are all current.					
d. Navigation Support	1. It is easy to understand where you are on this web site.					
	2. The use of panels, frames, or dedicated areas to facilitate navigation is very useful.					
	3. The site map, or overview is effective.					
e. Search Facilities	1. This site enables you to search on its content.					
	2. Searching on this site yields good (and relevant) results.					
	3. The way searches are structured is very useful.					
f. Attractiveness	1. This web site is attractive.					

g. Interaction Opportunities	2. This web site is not annoying.								
	3. Animations, blinking effects, music, banners, etc. support the purpose of this web site.								
	1. There are ample opportunities to interact with this web site.								
h. Error Reduction/ Recovery	2. Interactions that this web site allows do really support its purpose.								
	3. The types of interactions permitted are innovative.								
	1. The design of this web site reduces mistakes in user understanding of content.								
i. Help/ Documentation	2. This web site does not allow a user to undo an error.								
	3. It is easy to make navigation errors in this site.								
	1. You are able to find a help page, or help-related page quite easily.								
j. Reliability	2. Users will need assistance in using this web site.								
	3. The help documentation is relevant and effective to the workings of this web site.								
	1. The server of this web site is down or busy making browsing very slow or not effective.								
	2. There have been no interruptions in using this web site.								
	3. I have rarely found problems accessing this web site.								

Comments and Suggestions:

THANK YOU!

Respondent's Signature

CURRICULUM VITAE

The Curriculum Vitae shall reflect a listing of the personal information, educational background, expertise, trainings, and seminars attended by the researcher, and other pertinent information.

It should follow a single space format for every item and double space format between parts.

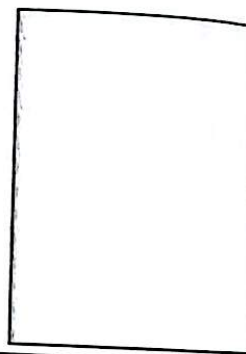
A passport size, colored picture with plain white background shall be placed at the upper right hand corner of the page. The researchers must wear business attire for the photograph.

In case of group research, each member of the group shall have a separate curriculum vitae.

Contents: Name, Permanent Address, Date of Birth, Place of Birth, Civil Status, Contact Number, E-mail Address, Educational Background, Achievements, Trainings/Seminars, Work Experiences, Awards Received, Special Skills/ Talents

(Sample Curriculum Vitae)

JOANNA MUNAR MARZAN
San Fabian, Pangasinan
2433 Philippines
Contact No: 0923-012-3456
E-mail: jmm@yahoo.com



Educational Background

2012 – present

Bachelor of Science in Computer Science
Don Mariano Marcos Memorial State University
Consolacion, Agoo, La Union
Thesis: "Mobile Application Using Java and HTML5"

2004- 2008

Catuday National High School
Catuday, Bolinao, Pangasinan

1998 - 2004

Catuday Elementary School
Catuday, Bolinao, Pangasinan

Oct 2008-Oct 2011

Customer Service Representative (CSR)
Caltex Gas Station
Paco, Maynila

Work Experiences

March 2015

Leadership Award
CSBO, DMMMSU-SLUC
Agoo, La Union

Achievements

March 2015

1st CCS Research Conference on ICT (RCICT 2015)
College of Computer Science
DMMMSU- SLUC
Agoo, La Union

January 2015

Mobile App Development using HTML5 and Phone Gap
College of Computer Science
DMMMSU- SLUC
Agoo, La Union

November 2014

Empowering CS/IT Research and Instruction
College of Computer Science
DMMMSU- SLUC
Agoo, La Union

Special Skills and Talents

- Programming skills in Visual Basic and Java
- Graphics editing using Adobe Photoshop
- Document preparation and presentation using Microsoft Office Applications

Certified true and correct.

Signature over printed name